

$\eta_g : \eta_s$	HR@50	NDCG@50	$\lambda_g : \lambda_s$	HR@50	NDCG@50	$\mu_{item} : \mu_{user}$	HR@50	NDCG@50
1:0	0.467	0.131	1:0	0.471	0.132	1:0	0.474	0.131
0:1	0.471	0.132	0:1	0.478	0.134	0:1	0.477	0.114
2:1	0.499	0.134	2:1	0.515	0.147	2:1	0.481	0.136
1:2	0.486	0.136	1:2	0.512	0.144	1:2	0.501	0.142
1:1	0.522	0.146	1:1	0.522	0.146	1:1	0.522	0.146

Table 7: Results using different hyper-parameters.

4.8 Impacts of Hyper-parameters and Ablation Study

The three sets of balance hyper-parameters η , λ and μ are used to balance the effects of representations from graph and sequence views, graph-view and sequence-view loss functions, and user and item loss functions, respectively.

Tab. 7 reports the performance of PC-Attack when attacking WRMF using different balance hyper-parameters. We can observe that the change of balance hyper-parameters affect the performance of PC-Attack. When setting equal values to all balance hyper-parameters, the attack performance is the best.

Besides, the first two rows and the last row in Tab. 7 can be viewed as three ablation experiments: (1) only use graph-view representation, graph-view loss and item loss, (2) only use sequence-view representation, sequence-view loss and sequence loss, and (3) the default PC-Attack that uses all parts. From Tab. 7, we can see that PC-Attack performs best when all parts are present and removing any of them will degrade the attack performance. Therefore, we can conclude that each component in PC-Attack indeed contributes to its overall attack performance.

5 Conclusion

In this paper, we study practical shilling attacks. We analyze the limitations of existing works and design a new framework PC-Attack that transfers the knowledge to attack the victim RS on incomplete target dataset. Experimental results demonstrate the superiority of PC-Attack. In the future, we plan to introduce more self-supervised learning tasks so that PC-Attack can get more supervision signals and better capture the transferable RS knowledge.

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